You need:

- Screen: 1.28 inch TFT VER 1.0, 240x240 , IC:GC9A01
- ESP32: ESP32U wroom
    + I got both of these as 2-packs for ~$30 on amazon.
- Breadboard and female-female dupont wires for testing
- A free OpenSky account

Step 1

After getting the parts, you're going to need Arduino IDE (ESP-IDF is also an option, but I didn't use it). I'd suggest not trying to solder anything/do anything with the board/screen yet, but you do you. You'll also need to download drivers for the ESP32 board (step 6)

In Arduino IDE:

Step 2

Tools -> Board -> Boards Manager. In the search bar on the left, type "ESP32" and install the one by Espressif Systems. This may take a minute or two

Step 3

After it's installed: Go to Tools -> Board -> ESP32 -> and select the one called "ESP32 Dev Module". From my understanding, this should work for the majority of ESP32 boards.

Step 4

Next, we need libraries. Go to Tools -> Manage Libraries. In the search bar on the left, search for and install "ArduinoJson" and "TFT_eSPI"

On your Computer:

Step 5

Go to GeGe's github link and copy "code V8" and paste it into Arduino IDE. You can delete/paste over the code that pops up when you open Arduino IDE. You shouldn't need to change anything about the code. We'll get to inputting your WIFI and coordinates later. The one thing you can change now is on line 12: If you want imperial instead of metric, change "true" to "false"

Step 6

You'll now need to install drivers for your computer to recognize the ESP32. For mine (and it seems like lots of ESP32's), CP210x drivers worked. Google "CP210x Driver" and go to silabs.com. Go to Downloads, then download the appropriate one. a. If you're on windows, Unzip the file, then open Device Manager. You should see a device under "Ports" or "Other Devices" when you plug your ESP32 into your computer. Verify you've got the correct one by watching Device Manager and plugging/unplugging the ESP32. b. Once you've got your ESP32, right click and select "Update Driver," then "Browse my computer." Select the unzipped folder and hit "OK"

Step 7

Now, under Tools -> Port, you should be able to select COM3. Your computer and the ESP32 are now talking

Step 8

Now we need to configure the TFT library so it knows what pins we're using. Find the folder your Arduino IDE is in (Mines in the Documents folder).
Go to Arduino -> Libraries -> TFT_eSPI -> Open "User_Setup.h" with notepad or another text editor. If you're using the same board as GeGe (GC9A01), do the following:

a) Section 1, Line 65: Delete the "//" before "#define GC9A01_DRIVER"
b) Section 1, Line 88: Delete the "//" before "#define TFT_WIDTH 240 // ST7789 240 x 240 and 240 x 320"
c) Section 1, Line 93: Delete the "//" before "#define TFT_HEIGHT 240 // GC9A01 240 x 240"
d) Section 2, Lines 212 - 217 : Delete the "//" before each of those 6 lines. The numbers after each should be:

- TFT_MISO -1
- TFT_MOSI 23,
- TFT_SCLK 18,
- TFT_CS 15,
- TFT_DC 2,
- TFT_RST 4.

I didn't have a backlight pin on my GC9A01 screen, so I didn't have to worry about that, but you may need to. I don't know, I'm not a programmer.

Quick note on step 8: The "//" makes comments in programming. If something is a comment, it's ignored by the code. The person who wrote this added a ton of options and deleting the "//" is basically us turning on that option, as the code won't ignore that line anymore. Also, if you're not using GC9A01, you're going to uncomment a different driver. If you're not using the same pins as GeGe, you need to match up the numbers accordingly.

Step 9

Save and close User_Setup.h. Then close Arduino IDE and open it again to make sure it loads everything correctly.

Back in Arduino IDE:

Step 10

Finally, our code/IDE should be ready. In the top left (blue buttons), hit "Verify" a. This is where you may need to troubleshoot. If errors pop up, google the problem and hopefully you'll get an answer. If not, leave a comment here and hopefully someone can help you.

Step 11

If it worked and your ESP32 is plugged in and recognized by the computer (Tools -> Board should be ESP32 Dev Module and Tools -> Port should be "COM3"), you're now ready to send the code to the ESP32. a. Hit the 2nd blue button, "Upload" b. In the bottom box words will start popping up. It'll say, "Connecting............" YOU NOW NEED TO PUSH AND HOLD THE "BOOT" BUTTON ON YOUR ESP32. Hold that button down until you see, "Connected to ESP32 on COM3." Shortly after, you'll see "Writing at #######" with a progress bar that will slowly fill up.

Step 12

Unplug the ESP32 from your computer. Now, we can wire the screen and ESP32 together. Again, I'd suggest NOT soldering yet. If you have a breadboard and female-female dupont wires, use them. If not do what you must. Follow GeGe's pinout on the github page. It should be self explanatory, and all the connections should be labled on each component.

Step 13

After everything is connected, you can plug the USB cord back into the ESP32. You can use an outlet instead of your computer at this point. The ESP32 just needs power. The screen should pop up with a few things, eventually settling on "Portal Active 192.168.4.1"

Step 14

Now, on a phone/tablet/computer, you should see a wifi network called, "DeskRadar-Setup". Connect to this, then go to 192.168.4.1 in your browser. Your device will say "Network has no internet", this is fine and normal. If the 192.168.4.1 page loads, you'll see some boxes to fill. This is where you'll select your Wifi, input the password, and input your latitude and longitude.

Step 15

At the bottom, you'll see the OpenSky information. To get that, you need to go to opensky-network.org and create an account. You'll get an email to verify your account. After you click on this link, you'll be met with the "Account" page on OpenSky. a. On the bottom right, you'll see the "API Client" box. Click the "Create Credential" box. A file called "credentials.json" will download. Open this and you'll see {"clientId":"USERNAME-api-client","clientSecret":"RANDOM CHARACTERS"}. b. You're going to use the username (WITHOUT THE "-api-client" PART!!) and the secret key on the DeskRadar-Setup webpage. DO NOT use your email on the DeskRadar-Setup page. Use your username.

Step 16

Type "1234" into the top box for your device pin, then hit the Save and Reboot button at the bottom of the page. The pin is needed anytime you change settings. You can change the pin in Arduino IDE, Line 14 of the code. a. When the ESP32 is operating normally, you won't be able to access this 192.168.4.1 webpage. This is only available when the ESP32 is in "Settings Mode". To get back into Settings Mode, push and hold the "Boot" button on the ESP32

The ESP32 should now restart and the screen should show the radar!

After getting it running, I soldered the wires for a permanent connection, then put everything in the 3d printed case.